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Experimental Validation of Yaw Stability Control Strategies for Articulated Vehicle Combinations

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***Abstract*— Articulated Heavy Vehicles (AHVs) play a crucial role in today's transportation, offering significant commercial and environmental advantages. However, challenges like jackknifing and trailer swing in AHVs highlight the need for focused research. This paper introduces innovative yaw stability algorithms designed to tackle these concerns, employing advanced control allocation techniques, including distributed control allocation. A power loss minimization algorithm with four different methods to maintain yaw stability is tested with real test vehicles. In this framework, the algorithms ensure that control actions stay within predetermined safe limits, contributing significantly to the overall safety and efficiency of AHVs.**

I. Introduction

The popularity of Articulated Heavy Vehicles (AHVs) is growing due to their significant benefits for businesses and the environment [1], [2]. An AHV, composed of a truck or tractor unit along with any number of trailers and dollies, often faces issues like jackknifing and trailer swing, extensively studied by researchers [3]–[7]. Jackknifing, in particular, is a major cause of serious accidents involving AHVs, with over 6000 reported in the US in 2021 [8].

Furthermore, over-actuation in AHVs introduces an additional layer of complexity that directly impacts their dynamics, imposing challenges in effectively managing and coordinating excessive control signals. Control allocation is a widely adopted and well-studied approach for optimizing the distribution of actuator signals in heavy vehicles (e.g. [9]–[11]). However, limited number of publications are available on control allocation for multi-unit vehicles (e.g. [12]–[14]), highlighting the need for improvements in this area. In [12], [13], a centralized control allocation is proposed for the entire combination, while in [14], a distributed two-layered control allocation is presented. One layer manages the combination level, distributing the requested global forces and moments among the vehicle units, while another layer is responsible for allocating the forces sent by the first layer to each unit's different actuators. The distributed control allocation offers the advantages of high reconfigurability for different types and numbers of trailing units, easier interface standardization, and better computational efficiency, making it a suitable choice for this study.

To address energy consumption in over-actuated electric vehicles, researchers have explored various control allocation methods. Torinsson et al. achieved a 3.9% reduction in energy consumption through power loss minimization algorithms [15]. Similarly, Janardhanan et al. compared different control allocation algorithms and concluded that the power loss minimization is the most energy-efficient approach for single-unit trucks [16].

Moreover, ensuring crucial yaw stability in AHVs has led to increased interest in electronic stability controllers [17]. While previous studies have explored active control strategies at the unit level, necessitating control over individual actuators [12], [18], [19], our research introduces a novel distributed motion control strategy at the combination level with a primary focus on power loss minimization.

Using safe force limits for the control allocation, namely Safe Operating Envelope (SOE) [20], [21], is a strong method ensuring the stability of the vehicle combinations. Alternatively, the safe limits can be defined in side-slip angle, side-slip angle rate, lateral velocity, and yaw rate domains [22]–[24]. In addition to power loss minimization, for maintaining yaw stability, four different methods are proposed in this paper. Within this framework, control actions are constrained within predetermined safe limits, significantly contributing to the overall safety and efficiency of AHVs.

The paper is structured as follows: The next section details the specifications of the test vehicle and test tracks utilized. Following that, the functionality of the proposed control allocation for both combination and unit levels are described. Subsequently, the yaw stability algorithms integrated into the control allocation functions are explained. Finally, the real test results are presented for the implementation of the proposed approaches, followed by the concluding remarks.

II. Test Vehicle Specification

The test vehicle, shown in Fig. 1, consists of a 4x2 Designwerk Futuricum electric tractor (with 500 kW power generated by 4 identical electric motors connected to the rear axle with a single gear, built on a Volvo FH chassis) [25] and a conventional semitrailer with 3 axles. The first and third axles of the semitrailer remain lifted during tests due to the small payload (due to weight restrictions on the test track). In this setup, a total weight of 19.47 tons is distributed among different axles, with 3 tons of payload loaded on the semitrailer, as illustrated in Fig. 1. Both vehicle units are equipped with highly accurate OxTS RT3000 GNSS/INS (Global Navigation Satellite System/Inertial Navigation System) [26] for measuring global position, angular rates, accelerations, and side-slip angles. Additionally, the semitrailer is equipped with an articulation angle sensor.

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All functions in this study are developed using MATLAB® Simulink®. Vehicle motion control is performed via a dSPACE MicroAutoBox II [27] with a 0.001 second time step, connected to vehicle CAN channels, electric powertrain CAN channels, sensors, tractor and semitrailer EBS advanced engineering interfaces for individual wheel braking requests. All CAN signals are logged by using CAN loggers. A jackknifing protection setup, shown in Fig. 2, is installed on the tractor and semitrailer. Steel cables in this setup get stretched at around 55° articulation angle, preventing a catastrophic jackknifing where the tractor cab collides with the semitrailer.

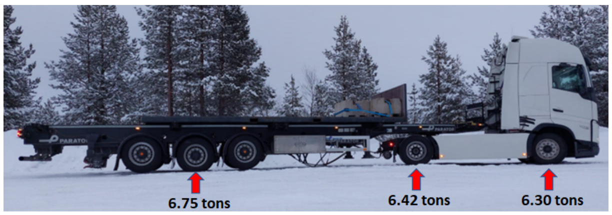

Fig. 1. Electric 4x2 tractor and conventional semitrailer

Fig. 2. Jackknifing protection cables shown under the semitrailer

III. Test Track Specification

Tests are performed at the Colmis Test Facility, Arjeplog, Sweden, as shown in Fig. 3. Brake-in-turn tests are performed on the circular lake track, following an approximately 130 m turning radius. The surface friction coefficient, μ, was measured using two test methods: driving the vehicle at highest possible speed in a circle, and fully braking all axles in a straight line. In both tests, the friction coefficient, μ, was observed approximately as 0.25 uniformly, a typical value for packed snow.

Fig. 3. Snow and ice circular tracks, Colmis Test Facility [28]

IV. Functional Description of Control Allocation

Vehicle motion control utilizes distributed control allocation, illustrated in Fig. 4, with a two-layered approach: the Combination-level Control Allocator (CCA), and the Unit-level Control Allocator (UCA). The CCA receives the global force request, F_x^{req}, and allocates it into different actuator types (electric motor and foundation brakes) for the tractor and semitrailer units. In this scenario, F_x^{req} is allocated to 4 different requests: $F_{x,e1}^{req}$, $F_{x,b1}^{req}$, $F_{x,e2}^{req}$, and $F_{x,b2}^{req}$. Here, subscripts e and b stand for electric motor and foundation brake, respectively, while subscripts 1 and 2 represent the first unit (tractor) and the second unit (semitrailer). The UCAs then distribute the received requests to various actuators within the respective units, denoted as $F_{x,e1,i}^{req}$, and $F_{x,b1,m}^{req}$ or $F_{x,e2,j}^{req}$, and $F_{x,b2,n}^{req}$, with subscripts i, m, j, and n representing the indices of the corresponding actuators.

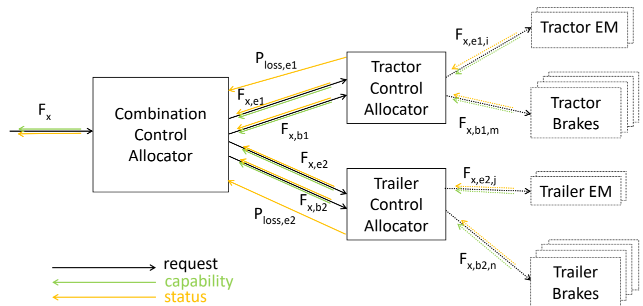

Fig. 4. Distributed functional architecture and signal interfaces

Fig. 4 depicts the longitudinal force allocation exclusively. Steering and yaw moment control are beyond the scope of this study. Following subsections will provide a detailed explanation of the CCA and UCA processes.

A. Combination-Level Control Allocation

The CCA employs a power loss minimization algorithm [15], [16]. The power losses, including copper, iron, windage, and friction losses in the electric machines, are approximately a second-order function of the motor torques at a specific motor speed. Utilizing an efficiency map of an electric motor (including the inverter losses), in multiplication with the other efficiencies (e.g. driveline and battery efficiencies), the total efficiency map can be calculated, representing efficiency from the battery to the wheels. This efficiency map is then used to calculate the approximated quadratic power loss function for each motor speed. Finally, this power loss function per each vehicle unit is sent as status signals, denoted as $P_{loss,e1}^{stat}$ and $P_{loss,e2}^{stat}$, from the UCAs to the CCA as shown in Fig. 4 and expressed as:

$$\begin{aligned} P_{loss,e1}^{stat} &= a_1 T_1^2 + b_1 T_1 + c_1 \\ P_{loss,e2}^{stat} &= a_2 T_2^2 + b_2 T_2 + c_2 \end{aligned} \tag{1}$$

Here, T_1 and T_2 represent the electric motor torques at each unit. $a_1, b_1, c_1, a_2, b_2, c_2$ are coefficients of the quadratic functions. For the positive and negative torques, two different curve fittings are done to enhance precision. The quadratic optimization problem addressed in the CCA is given as:

$$\begin{aligned} &\min_u \left(\frac{1}{2} u^T H u + g^T u \right) \\ &\text{s.t. } A_{eq} u = b_{eq}, \qquad u_{\min} \leq u \leq u_{\max} \end{aligned} \tag{2}$$

Here, $u_{\max}$ and $u_{\min}$ denote the upper and lower bounds for the allocated forces, representing the actuator capabilities per each unit (i.e. $F^{cap}_{x,e1}$, $F^{cap}_{x,b1}$, $F^{cap}_{x,e2}$, and $F^{cap}_{x,b2}$). The matrices H, A_{eq}, and the vectors g, b_{eq}, u are defined as follows:

$$
\begin{aligned}
&H = 2\begin{bmatrix} r^2a_1/k_1^2 & 0 & 0 & 0 \\ 0 & h_{brk} & 0 & 0 \\ 0 & 0 & r^2a_2/k_2^2 & 0 \\ 0 & 0 & 0 & h_{brk} \end{bmatrix} \\
&g^T = \begin{bmatrix} rb_1/k_1 & -V_{x,1} & rb_2/k_2 & -V_{x,2} \end{bmatrix} \\
&A_{eq} = \begin{bmatrix} 1 & 1 & \cos(\theta) & \cos(\theta) \end{bmatrix} \\
&b_{eq} = F^{req}_x, \quad u^T = \begin{bmatrix} F^{req}_{x,e1} & F^{req}_{x,b1} & F^{req}_{x,e2} & F^{req}_{x,b2} \end{bmatrix}
\end{aligned}
\tag{3}
$$

The equality constraint $A_{eq}u = b_{eq}$ ensures that the requested total longitudinal force, F^{req}_x, is fulfilled by the allocated forces to the units. The CCA communicates the time-varying longitudinal force capability, F^{cap}_x, to the upper-level controller to prevent excessive requests. F^{req}_x is then saturated according to F^{cap}_x, ensuring fulfillment of F^{req}_x without feasibility issues. Here, r, k_1, and k_2 denote the wheel radius, and gear ratios, converting unit force to motor torque in the linear and quadratic terms g and H. θ denotes the articulation angle. Vehicle speeds, $V_{x,1}$ and $V_{x,2}$, are used in the linear term g, as power losses of foundation brakes are simplified (by ignoring the wheel slips) as $V_{x,1}F^{req}_{x,b1}$ and $V_{x,2}F^{req}_{x,b2}$, which means that all the kinetic energy of the vehicle is dissipated as heat while the foundation brakes are used to slow down. Although the power loss terms for the foundation brakes are linear in u, a small value of $h_{brk} = 1e^{-5}$ is used as quadratic terms in the Hessian matrix H to ensure positive definiteness.

The quadratic optimization problem described is a generic problem for any tractor and semitrailer combination, which may both have electric motors. However, the test vehicle used for this study has a conventional semitrailer without an electric motor. Consequently, the third elements of $u_{\max}$ and $u_{\min}$ are set as 0, i.e. $u_{\min,3} = u_{\max,3} = 0$. This leads to a trivial solution in the power loss minimization problem (2): the optimal solution is to utilize the tractor electric motors for propulsion or regenerative braking until reaching the capability $F^{cap}_{x,e1}$ except at very small speeds, where the power loss in the electric motors are higher than the ones in the foundation brakes. Hence, before coming to a standstill at small speeds, braking is employed via the foundation brakes.

1) Coupling Force Enhancement Function: The g vector in (3) has equal linear terms for the tractor and semitrailer foundation brakes. This leads to equal service braking for both units once the electric motor capabilities are reached. For power loss minimization, any combination of the tractor and semitrailer regenerative braking can be justified. However, equal usage of foundation brakes does not necessarily reduce power consumption since all energy is dissipated as heat. Hence, prioritizing safety is essential. A safer approach for allocating unit foundation brake forces is aiming for equal friction utilization on the unit level. Thus, the total braking forces for the tractor and semitrailer should be proportional to the sum of their axle loads. In the case of the test vehicle used in this study (where $u_{min,3} = u_{max,3} = 0$ for the conventional semitrailer), equal friction utilizations per unit after reaching the electric motor capability, $u_{min,1}$, can be obtained with the additional equality constraint given in (4).

$$
\begin{aligned}
&A^{coupling}_{eq} u = b^{coupling}_{eq} \\
&\text{where } A^{coupling}_{eq} = \begin{bmatrix} \frac{1}{F^{stat}_{z,1}} & \frac{1}{F^{stat}_{z,1}} & 0 & -\frac{1}{F^{stat}_{z,2}} \end{bmatrix}
\end{aligned}
\tag{4}
$$

Here, $F^{stat}_{z,1}$ and $F^{stat}_{z,2}$ are the total axle loads for each unit. $b^{coupling}_{eq}$ is defined as a function of F^{req}_x as shown in Fig. 5. This equality constraint is activated only if $F^{req}_x < u_{min,1}$. Consequently, after reaching the tractor electric motor force capability, $u_{min,1}$, semitrailer foundation brakes are prioritized until reaching equal unit friction utilizations. Then, the foundation brakes of both units are used to fulfill the equal unit friction utilizations. Thanks to this coupling force enhancement function, excessive tractor braking and excessive coupling forces are avoided, leading to improved yaw stability.

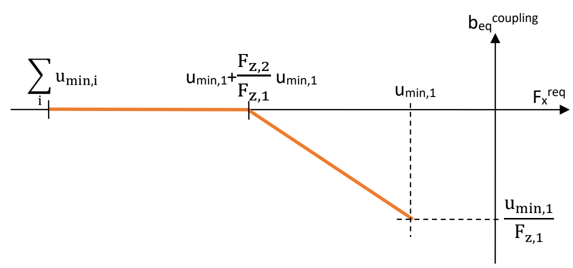

Fig. 5. $b^{coupling}_{eq}$ defined as a function of F^{req}_x for the coupling force enhancement

B. Unit-Level Control Allocation

UCAs, Unlike CCA, are simple rule-based controllers tailored to the test vehicle used in this study. In the following subsections, the control allocation for both vehicle units will be explained.

1) Control Allocation for Tractor: The total electric motor force request, $F^{req}_{x,e1}$, from the CCA is equally allocated among four electric motors of the tractor as $F^{req}_{x,e1,i}$, where $i \in \{1, 2, 3, 4\}$. From the power loss point of view, as all electric motors are identical with the same gear ratio, there is no need to run a power loss minimization algorithm as in (2) at the unit-level.

Until the front tires reach the friction utilization of the rear tires (which may already be regeneratively braked with electric motors), only the front foundation brakes ($F^{req}_{x,b1,1}$ for left and $F^{req}_{x,b1,2}$ for right) are equally used to fulfill the total brake request $F^{req}_{x,b1}$. Once the front and rear tires reach the same friction utilization, the rear foundation brakes ($F^{req}_{x,b1,3}$ for left and $F^{req}_{x,b1,4}$ for right) are used together with the front ones to ensure similar friction utilizations for both front and rear, based on given front and rear axle loads, i.e. $F^{stat}_{z,1f}$ and $F^{stat}_{z,1r}$. Refer to Algorithm 1 for the tractor control allocation.

2) Control Allocation for Semitrailer: Control allocation for the semitrailer is trivial. The left ($F^{req}_{x,b2,1}$) and right ($F^{req}_{x,b2,2}$) foundation brakes are equally used to fulfill the total brake request $F^{req}_{x,b2}$, as shown in Algorithm 2.

Algorithm 1 Tractor control allocation

Require: Input values for $F^{req}_{x,e1}$, $F^{req}_{x,b1}$, $F^{stat}_{z,1f}$, $F^{stat}_{z,1r}$
  $F^{req}_{x,e1,i} \leftarrow F^{req}_{x,e1}/4$, $i \in \{1,2,3,4\}$
  if $F^{req}_{x,b1} \geq F^{req}_{x,e1}\frac{F^{stat}_{z,1f}}{F^{stat}_{z,1r}}$ **then**
    $F^{req}_{x,b1,m} \leftarrow \frac{F^{req}_{x,b1}}{2}$ for $m \in \{1,2\}$ ▷ brake front wheels
    $F^{req}_{x,b1,m} \leftarrow 0$ for $m \in \{3,4\}$ ▷ don't brake rear wheels
  else ▷ brake front/rear proportional to normal loads
    Solve following equations for $F^{req}_{x,b1,m}$, $m \in \{1,2,3,4\}$
    $\frac{F^{req}_{x,b1,1}}{F^{stat}_{z,1f}} = \frac{F^{req}_{x,b1,3} + F^{req}_{x,e1}/2}{F^{stat}_{z,1r}}$
    $F^{req}_{x,b1,1} = F^{req}_{x,b1,2}$ ▷ brake right/left equally
    $F^{req}_{x,b1,3} = F^{req}_{x,b1,4}$ ▷ brake right/left equally
  end if
  return $F^{req}_{x,e1,i}$, $F^{req}_{x,b1,m}$, $i \in \{1,2,3,4\}$, $m \in \{1,2,3,4\}$

Algorithm 2 Semitrailer control allocation

Require: Input value for $F^{req}_{x,e2}$
  $F^{req}_{x,b1,n} \leftarrow \frac{F^{req}_{x,b2}}{2}$ for $n \in \{1,2\}$ ▷ brake right/left equally
  return $F^{req}_{x,b2,n}$, $n \in \{1,2\}$

V. Yaw Stability Algorithms

In this section, several algorithms designed to preserve the yaw stability of vehicle combinations are explained.

A. Reducing Longitudinal Force Capability Based on Lateral Force Estimation

The free-body diagrams of the tractor and semitrailer are depicted in Fig. 6. The equations of motion in the horizontal plane and compatibility equations are given in (5). These equations assume a zero yaw angular acceleration and a zero steering angle.

$$
\begin{aligned}
m_1 a_{x,1} &= F_{x,1} + F_{x,c} \\
m_2 a_{x,2} &= F_{x,2} - F_{x,c}\cos\theta - F_{y,c}\sin\theta \\
a_{x,1} &= a_{x,2}\cos\theta + a_{y,2}\sin\theta \\
m_1 a_{y,1} &= F_{y,1} - F_{y,c} \\
m_2 a_{y,2} &= F_{y,2} + F_{y,c}\cos(\theta) - F_{x,c}\sin\theta \\
a_{y,1} &= a_{y,2}\cos\theta - a_{x,2}\sin\theta \\
0 &= l_f F_{y,1f} - l_{r1} F_{y,1r} + F_{y,c} l_{v1} \\
F_{y,1f} + F_{y,1r} &= F_{y,1} \\
0 &= l_{v2}(F_{y,c}\cos\theta - F_{x,c}\sin\theta) - l_{r2} F_{y,2}
\end{aligned}
\tag{5}
$$

The set of 9 equations presented in (5) can be solved for 9 unknowns, which includes the longitudinal and lateral coupling forces, $F_{x,c}$, $F_{y,c}$; longitudinal and lateral accelerations of the semitrailer, $a_{x,2}$, $a_{y,2}$; lateral forces, $F_{y,1}$, $F_{y,2}$, $F_{y,1f}$, $F_{y,1r}$; and the longitudinal force applied by the semitrailer actuators, $F_{x,2}$. It is assumed that the masses of the tractor and semitrailer, m_1, m_2; longitudinal and lateral accelerations of the tractor, $a_{x,1}$, $a_{y,1}$; articulation angle, θ; and the longitudinal force applied by the tractor actuators, $F_{x,1}$ are known. The parameters used in the lateral force estimation are given in Table I.

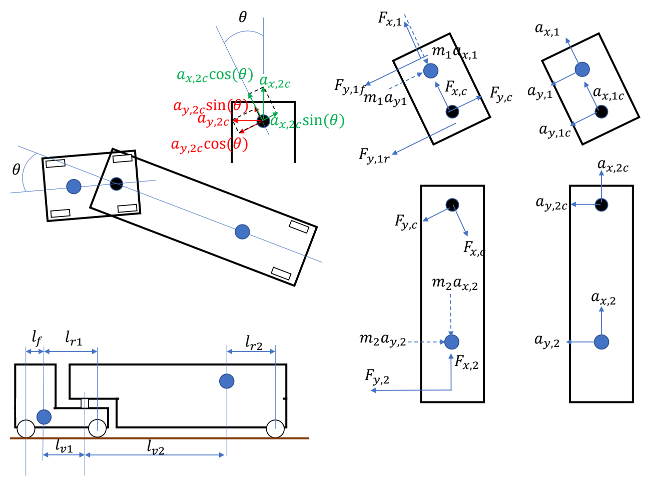

Fig. 6. Free body diagrams for the tractor and semitrailer combination

Algorithm 3 Reducing longitudinal force capability based on lateral force estimation

Require: Input values for $F^{est}_{y,1f}$, $F^{est}_{y,1r}$, $F^{est}_{y,2}$, C, μ, $u_{max,1}$, $u_{min,1}$, $u_{min,2}$, $u_{min,4}$, $F^{stat}_{z,1f}$, $F^{stat}_{z,1r}$, $F^{stat}_{z,2}$
  $F^{lim}_{x,1r} \leftarrow \sqrt{\max(\mu^2 C^2 F^{stat^2}_{z,1r} - F^{est^2}_{y,1r}, 0)}$
  $F^{lim}_{x,1f} \leftarrow \sqrt{\max(\mu^2 C^2 F^{stat^2}_{z,1f} - F^{est^2}_{y,1f}, 0)}$
  $F^{lim}_{x,2} \leftarrow \sqrt{\max(\mu^2 C^2, F^{stat^2}_{z,2}, -F^{est^2}_{y,2}, 0)}$
  $u^{modified}_{max,1} \leftarrow \min(F^{lim}_{x,1r}, u_{max,1})$
  $u^{modified}_{min,1} \leftarrow \max(-F^{lim}_{x,1r}, u_{min,1})$
  $u^{modified}_{min,2} \leftarrow \max(-F^{lim}_{x,1f} - F^{lim}_{x,1r} - u^{modified}_{min,1}, u_{min,2})$
  $u^{modified}_{min,4} \leftarrow \max(-F^{lim}_{x,2}, u_{min,4})$
  return $u^{modified}_{max,1}$, $u^{modified}_{min,1}$, $u^{modified}_{min,2}$, $u^{modified}_{min,4}$

TABLE I

PARAMETERS USED IN LATERAL FORCE ESTIMATION

Param.	Explanation	Value	Unit
m_1	tractor mass	9000	kg
m_2	semitrailer mass	10472	kg
l_f	distance from tractor CoG to front axle	1.35	m
l_{r1}	distance from tractor CoG to rear axle	2.45	m
l_{r2}	distance from trailer CoG to trailer axle	2.74	m
l_{v1}	distance from tractor CoG to coupling	1.87	m
l_{v2}	distance from trailer CoG to coupling	4.96	m

The estimated lateral axle forces of both units, $F^{est}_{y,1f}$, $F^{est}_{y,1r}$, $F^{est}_{y,2}$, can constrain the longitudinal forces applied by the actuators within the friction circle [29]. To enhance the algorithm's robustness against uncertain parameters, a downsized friction circle reduced by a factor of C is used. Here, C is set to 0.8, implying that axle forces are saturated at

80% of their friction circles. The proposed method is shown in Algorithm 3, modifying the upper and lower bounds (u_{max} and u_{min}) for allocated forces per unit and actuator type.

B. Safe Operating Envelope

A Safe Operating Envelope (SOE) is a set of lower and upper limits on actuator (braking or propulsion) requests, expressed as functions of vehicle states (e.g., lateral acceleration) and environmental parameters (e.g., road friction) [20]. SOEs for a variety of vehicle and environmental parameters are obtained in [21]. The safe limits of the actuators can be functions of each other, hence they can be written as in (6), rather than being confined to simple box constraints such as ($u_{min} \leq u \leq u_{max}$).

$$A_{in} u \leq b_{in} \tag{6}$$

The inequality constraint (6) can then be solved together with the quadratic optimization problem as in (2). The SOE used in this study is shown in Fig. 7, depicting eight different envelopes represented by different colors for eight different normalized lateral accelerations, c_y, which is defined as:

$$c_y = \frac{a_y}{\mu g} \tag{7}$$

where a_y is the lateral acceleration of the tractor, and g is the gravitational acceleration. The SOE for a given c_y is defined as the area in between the two lines of the same color.

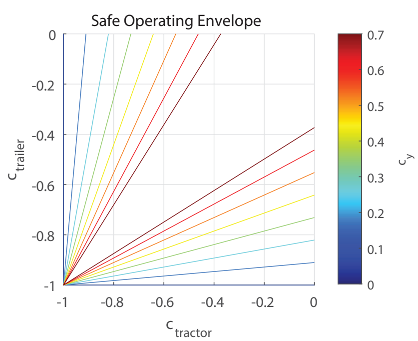

Fig. 7. Safe operating envelope (safe limits of tractor drive axle friction utilization, $c_{tractor}$, and semitrailer axle friction utilization, $c_{trailer}$) for braking with a variety of normalized lateral accelerations, c_y

The friction utilization for the tractor drive axle, $c_{tractor}$ and the friction utilization for the semitrailer axle, $c_{trailer}$ are represented as:

$$\begin{aligned} c_{tractor} &= \frac{F_{x,e1}^{req} + F_{x,b1,3}^{req} + F_{x,b1,4}^{req}}{\mu F_{z,1r}^{stat}} \\ c_{trailer} &= \frac{F_{x,e2}^{req} + F_{x,b2,1}^{req} + F_{x,b2,2}^{req}}{\mu F_{z,2}^{stat}} \end{aligned} \tag{8}$$

where $F_{z,2}^{stat}$ represents the semitrailer axle load. It should be noted that the semitrailer electric motor force, $F_{x,e2}^{req}$, is always zero, as the tested semitrailer is conventional. $c_{tractor}$ and $c_{trailer}$ include six input requests: $F_{x,e1}^{req}$ and $F_{x,e2}^{req}$ which are first and third elements of the u vector of the quadratic optimization problem given in (2). In this particular application, $F_{x,b2,1}^{req} + F_{x,b2,2}^{req}$ is equal to $F_{x,b2}^{req}$ (the fourth element of the u vector), as explained in Section IV-B.2. $F_{x,b1,3}^{req}$ and $F_{x,b1,4}^{req}$, on the other hand, involve input requests from the tractor UCA, as explained in Section IV-B.1. The status signals of the tractor wheel braking force quantities, $F_{x,b1,m}^{stat}$ for $m \in \{1, 2, 3, 4\}$, are sent from the tractor UCA to the CCA, as shown in Fig. 4. Hence, the CCA has the information of the ratio $(F_{x,b1,3}^{req} + F_{x,b1,4}^{req})/F_{x,b1}^{req}$ from the previous time step. This implies that $F_{x,b1,3}^{req}$ and $F_{x,b1,4}^{req}$ in (8) can be expressed as a function of $F_{x,b1}^{req}$ (the second element of the u vector) with a one-time-step delay. In conclusion, all terms given in (8) can be expressed as functions of the u vector, and the limits of the SOE shown in Fig. 7 can be written as in (6).

In this study, a modified version of the SOE derived in [21] is used to ensure yaw stability. For larger total brake force requests, the SOE gets tighter along the diagonal (drawn from the upper right corner to the lower left corner) in Fig. 7. This diagonal represents equal friction utilization for both the tractor and semitrailer, representing the safest allocation. Furthermore, the SOE gets tighter for higher c_y values. All these aspects of SOE ensure a safer control allocation for higher brake force requests or higher lateral accelerations. As the tractor and semitrailer brake forces are already limited depending on each other by the SOE, the coupling force enhancement function explained in Section IV-A.1 is disabled when SOE is used, to avoid infeasibilities during the solution of the optimization problem given in (2).

C. Reducing Longitudinal Force Capability with High Longitudinal Slip Detection

A simple approach to avoid yaw instabilities due to excessive wheel forces is reducing the longitudinal force capability after detecting high longitudinal slip on the driven wheels during high lateral accelerations, as shown in Algorithm 4. The same generic algorithm is used also for Section V-D with minor differences. In the scope of detection with high longitudinal slip, the input S_1 as given in Algorithm 4 depicts the highest longitudinal slip of the tractor driven wheels, $S_{x,1}$, and S_1^{lim} as given in Algorithm 4 depicts the longitudinal slip limit, $S_{x,1}^{lim} = 0.1$, for detecting the high slip. The lateral acceleration limit where the algorithm is activated, a_y^{lim}, is selected as 1 m/s^2, and the capability reduction factor, γ, is selected as 0.5. The algorithm returns the modified minimum and maximum electric motor force capabilities for the tractor, $u_{min,1}^{modified}$, $u_{max,1}^{modified}$. Unlike two previous proactive methods, this reactive method imposes force limitation after some states such as lateral acceleration and longitudinal slips exceed predefined limits.

D. Reducing Longitudinal Force Capability with High Lateral Slip Detection

An alternative method to detect potential yaw instability is using the lateral slip at the drive axle of the tractor. In the scope of detection with high lateral slip, the input S_1 as given in Algorithm 4 depicts the side-slip angle of the tractor drive axle, $S_{y,1}$; and S_1^{lim} as given in Algorithm 4 depicts the lateral slip limit, $S_{y,1}^{lim} = 2°$, for detecting the high slip. Hence, lateral slips are expressed as side-slip angles. Similar to the previous method shown in Section V-C, this method is also reactive.

Algorithm 4 Reducing longitudinal force capabilities with high slip detection

Require: Input value for S_1, S_1^{lim}, $u_{min,1}$, $u_{max,1}$, a_y, a_y^{lim}, γ, F_x^{req}, F_x^{lim}
Initialize: $capabilityReduced \leftarrow$ false
while $|F_x^{req}| < F_x^{lim}$ **do** ▷ while force request is small
  $u_{min,1}^{modified} \leftarrow u_{min,1}$ ▷ reset modified the capabilities
  $u_{max,1}^{modified} \leftarrow u_{max,1}$
  $capabilityReduced \leftarrow$ false
end while
if $|S_1| > S_1^{lim}$ & $|a_y| > a_y^{lim}$ & not($capabilityReduced$) **then**
▷ if there is high slip and high lateral acceleration
  $u_{min,1}^{modified} \leftarrow \gamma u_{min,1}$ ▷ reduce the capabilities
  $u_{max,1}^{modified} \leftarrow \gamma u_{max,1}$
  $capabilityReduced \leftarrow$ true
end if
return $u_{min,1}^{modified}$, $u_{max,1}^{modified}$

VI. Test Results

In this section, the yaw stability algorithms explained in Section V are used in real tests, and their results are presented. Brake-in-turn maneuvers may involve both high longitudinal and lateral accelerations, requiring high longitudinal and lateral forces. A combination of the high forces may lead to tire force saturation and consequently yaw instabilities such as jackknifing.

Several different yaw stability algorithms are tested with the CCA in brake-in-turn tests. In all tests, the vehicle combination is braked from a precise speed of 45 km/h. For the test track with approximately 130 m turning radius, this speed corresponds to approximately 1.2 m/s^2 lateral acceleration. The surface friction, μ, is measured as approximately 0.25. The slip controller of the electric motor and electronic stability controller are turned off (still no back-spinning allowed for the wheels) in all brake-in-turn tests to challenge each safety algorithm without getting help from these functions, and aiming to examine the worst-case scenarios. In all tests, a global force, F_x^{req}, of -15750 N (equal to $-\mu F_{z,1r}^{stat}$) is requested. During braking, the driver keeps the steering angle constant without trying to stabilize the vehicle.

A. No Yaw Stability Algorithm

When no yaw stability algorithm is used, the CCA allocates all the required force to the tractor electric motor as shown in Fig. 8. Braking regeneratively (indicated by dashed lines and an asterisk) only on the tractor drive axle results in the longitudinal and lateral slips, and yaw rate of the tractor to increase, eventually leading to jackknifing. When the articulation angle, θ, reaches to 55°, the jackknife protection cables get stretched, preventing catastrophic jackknifing. Then the driver takes over the control.

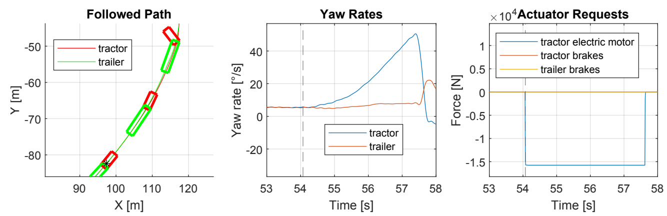

Fig. 8. State plots when no yaw stability algorithm is used

B. Reducing Longitudinal Force Capability Based on Lateral Force Estimation

Estimating the lateral forces on each axle and limiting the longitudinal force capabilities to 80% of the friction circle, leads to safe braking without jackknifing, as shown in Fig. 9. Tractor drive axle normalized forces (with respect to the normal load) are saturated at 0.2 (which is 80% of the designated friction coefficient, $\mu = 0.25$). When these forces are saturated, then the global force request is fulfilled mostly by supplementary semitrailer braking, and some tractor braking. Just before coming to a standstill, braking is only performed by foundation brakes as explained in Section IV-A for small speeds.

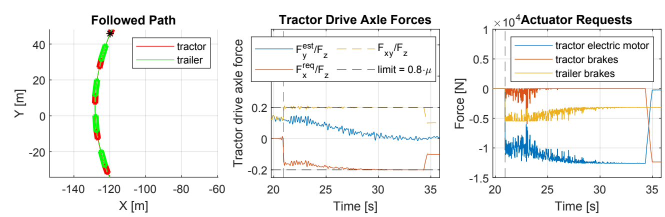

Fig. 9. State plots when longitudinal force capabilities are reduced based on lateral force estimation

C. Safe Operating Envelope

When using an SOE to ensure stability, as explained in Section V-B, the tractor is regeneratively braked with the electric motors to a certain extent, shown in Fig. 10. The total braking force request is then fulfilled by supplementary semitrailer braking. In the second plot of Fig. 10, SOEs for every second after braking are shown with 2 dashed lines with varying colors. Additionally, the allocated friction utilizations for the tractor drive axle and semitrailer axle ($c_{tractor}$ and $c_{trailer}$) are indicated with scattered dots. At the very start of the braking, SOE is tighter (shown in blue) due to high lateral acceleration, hence the tractor regenerative braking is supplemented with the semitrailer foundation brakes. After slowing down, lateral acceleration drops, and the SOE gets larger (shown in red). Thus, semitrailer braking is no longer required and the braking is fully performed by regenerative means via the tractor electric motors. At very small speeds, before coming to a standstill, braking is only performed by foundation brakes as explained in Section IV-A. Control allocation with SOE ensures safe braking, bringing the combination to a standstill without jackknifing.

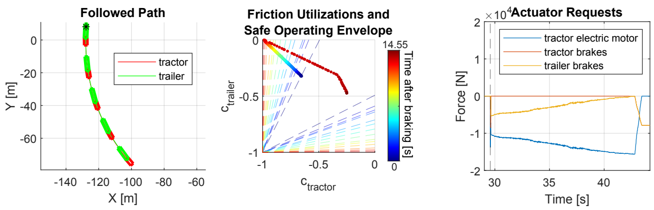

Fig. 10. State plots when safe operating envelope is used

D. Reducing Longitudinal Force Capability with High Longitudinal Slip Detection

Reducing the longitudinal force capability when a high longitudinal slip and high lateral acceleration are detected, as described in Section V-C, ensures safe braking, as illustrated in Fig. 11. The start of braking is shown with vertical dashed lines in the second and third plots, and with an asterisk in the path plot. The combination starts braking only via tractor electric motors. However, after 0.65 seconds, the longitudinal slip of the left driven wheel of the tractor exceeds -0.1. Then the algorithm described in Section V-C reduces the tractor electric motor force capabilities by 50%. This moment is shown with a second dashed line in the second and third plots, and with another asterisk in the path plot. After reducing the electric motor force capability, braking is supplemented by tractor and semitrailer foundation brakes. Consequently, the longitudinal wheel slip decreases and the combination can brake safely.

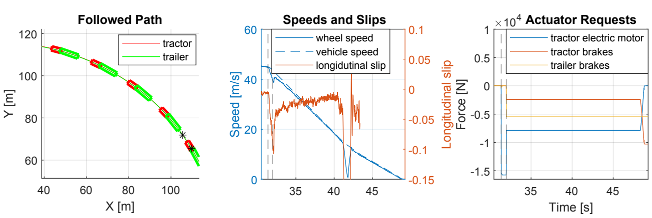

Fig. 11. State plots when the longitudinal force capability is reduced with high longitudinal slip detection

E. Reducing Longitudinal Force Capability with High Lateral Slip Detection

A brake-in-turn maneuver when the longitudinal force capability is decreased due to high lateral slip and high lateral acceleration results in safe braking, as shown in Fig. 12. Similar to the previous figure, the first vertical dashed lines and first asterisk denote the start of braking. The second vertical dashed lines and asterisk denote when the high lateral slip is detected, which is approximately 1 second after the braking starts. After detection of high (absolute value of) lateral slip, tractor electric motor force capability is reduced by 50% and braking is supplemented with tractor and semitrailer foundation brakes. Consequently, (absolute value of) the lateral slip of the tractor decreases and the combination comes to a standstill in a stable way.

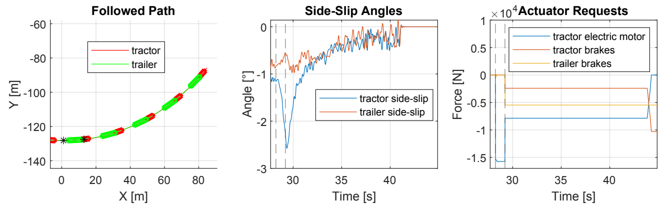

Fig. 12. State plots when the longitudinal force capability is reduced with high lateral slip detection

VII. Conclusion

In this study, control allocation with power loss minimization and a distributed functional architecture are introduced. Following this, four different methods designed for use with combination-level control allocator to maintain yaw stability are explained. Finally, these methods are tested with a real vehicle combination, and their performances are discussed.

The first two methods, which involve reducing the longitudinal force capability based on lateral force estimation and using a safe operating envelope, are proactive approaches aiming to prevent instability before it happens. Conversely, the other two methods, focused on reducing the force capability based on high lateral or longitudinal slip detection, are reactive, meaning that they aim to bring the vehicle back to a stable operating point after a potential instability starts. The proactive methods keep the yaw rates under 6 °/s, whereas the reactive methods become active after yaw rates grow up to 8 °/s during the same maneuver.

The reactive methods should be studied in detail and simulated with various scenarios to identify the proper slip limits and capability reduction ratios. Instead of a stepwise reduction in the capabilities, gradual ramping down should also be considered as future work. In this case, capabilities would be gradually ramped down until the slips decrease, potentially leading to more energy-efficient driving. The proactive methods, on the other hand, adopt a model-based approach and should already perform well under a wide variety of conditions. SOE, for example, can be obtained by running a high-fidelity vehicle model in different scenarios as shown in [21], and thus the method utilizing SOE should mostly result in safe maneuvers. The other proactive method involving lateral force estimation should ensure a safe maneuver as long as the estimation is done accurately and the operating conditions are known with high accuracy. Both proactive methods require the knowledge of friction which can be estimated, but the rest of the required information are rather simple: vehicle parameters and IMU measurements.

All or a subset of the methods shown in this study can be used in combination-level control allocation to ensure a safe allocation. All methods are possible to integrate with the base control allocation strategy (power loss minimization) shown in Section IV-A in the combination level, without involving unit-level allocation (such as braking an individual wheel to generate yaw moment). Proposed methods can be implemented with either inequality constraints or box constraints on the capabilities together with (2). These algorithms are designed to maintain vehicle stability, while (2) continuously seeking the most energy-efficient allocation. If these algorithms fail to ensure a stable allocation, then electronic stability controllers of the units can kick in and actively stabilize the vehicle.

All methods shown in this study involve limiting the forces even though some involve instability detection based on high slip. Reactive methods involving slip-based detection show a promising direction in the research. Instead of only relying on force-based limitations, slip-based methods should be also investigated in future work, such as limiting the longitudinal slip of the actuators based on the lateral slip.

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